Functional Design Document (FDD) – Test-A-Form: E-Form Data Capture System

1. Document Control

- Project Name: Test-A-Form – E-Form System
- Module Name: Data Collection & Validation
- Author: Spoorthi M (spoorthimaheshbhat@gmail.com)
- Date: 05-08-2025
- Version: V-2.0

---

2. Purpose

This document defines the functional design of the Test-A-Form e-form system, which enables users to input, validate, submit, and optionally edit responses through a structured interface.

The document serves as a reference for:

- Requirement understanding
- System behavior validation
- Test case design
- Automation implementation

---

3. Scope

This document outlines the functional features of the Test-A-Form e-form system, designed to simulate a real-world data collection application.

The system supports user interaction through a structured form interface, enabling data capture, validation, submission, and post-submission response management.

In Scope

- Accessing the form via a publicly available URL
- Display and interaction with various input elements (text fields, multiple-choice, dropdowns, checkboxes, date/time inputs)
- Validation of user inputs including:
    - Mandatory field checks
    - Email format validation
    - Numeric range constraints
- Multi-section navigation within the form
- Submission of user responses

- Display of confirmation message upon successful submission
- Storage of submitted responses in a linked Google Sheet
- Post-submission functionality allowing users to edit their responses via “Edit your response” option
- Pre-filling of previously submitted data during edit
- Re-submission and update of modified responses

Out of Scope

- API-level validations
- Backend/database validation beyond Google Sheets
- Authentication-based access control
- Performance and load testing
- Security testing (e.g., CAPTCHA, bot prevention)

---

4. Functional Overview

The system is a publicly accessible e-form built using Google Forms that simulates real-world data collection.

It facilitates structured data capture with built-in validation, submission handling, and post-submission response management capabilities.

Users can:

1. Access the form
2. Enter required and optional data
3. Navigate across sections
4. Submit the form
5. View confirmation
6. Optionally edit their response

---

5. Stakeholders:

- End Users: Individuals submitting responses
- Tester/QA: Validates system functionality
- Business Analyst: Defines requirements and validation logic
- System Owner: Maintains form and reviews responses

---

6. End-to-End Workflow

User → Access Form (Public URL)→ Enter Data → Validation → Navigate Sections → Submit → Confirmation Message → Edit Response (Optional) → Update Submission

---

7. Functional Requirements

List and description of individual requirements.

ID	Requirement Description	Priority
FR-01	System must open the Google Form using the provided public URL.	High
FR-02	System must display all form elements (input fields, dropdowns, checkboxes) and ensure they are interactable.	High
FR-03	System must validate mandatory fields	High
FR-04	System must validate email format	High
FR-05	System must validate numeric range (1–10)	High
FR-06	System must allow navigation between sections	Medium
FR-07	System must submit form upon valid input	High
FR-08	System must display appropriate validation messages upon incorrect/missing test data input into each form field.	Medium
FR-09	System must verify that a confirmation message is displayed after form submission.	High
FR-10	System must validate that a new response has been recorded in the linked Google Sheet.	High
FR-11	System must allow users to edit submitted responses	High
FR-12	System must pre-fill previously entered data during edit	High
FR-13	System must update stored response upon resubmission	High
FR-14	System should maintain consistent layout and alignment of form elements across sections.	Low
FR-15	System should display descriptive labels and helper text for each field to guide the user.	Low
FR-16	System should initialize all input fields in a default unfilled state upon first load.	Low
FR-17	System should allow users to leave non-mandatory fields blank without impacting submission.	Low
FR-18	System should allow users to scroll through the form without UI distortion or overlap.	Low
FR-19	System should function correctly on standard desktop browsers (e.g., Chrome, Edge).	Medium
FR-20	System should be accessible and function correctly on mobile devices, ensuring all form elements are visible, readable, and interactable on smaller screens.	Medium
FR-21	System should support usage on mobile devices, ensuring proper rendering of UI elements, smooth navigation between sections, and accurate data input and submission.	Medium
FR-22	System should support touch-based interactions on mobile devices, including selection of dropdowns, checkboxes, and date/time inputs without usability issues.	Low

8. Assumptions

- The e-form is publicly accessible and does not require user authentication for submission.
- The structure, fields, and validation rules of the form remain consistent during its usage.
- The linked Google Sheet is properly configured to capture and store all submitted responses accurately.
- Users have access to a stable internet connection and a supported web browser.
- The system relies on standard Google Forms functionality for validation and data handling.
- No CAPTCHA, authentication, or bot protection mechanisms are enabled on the form.
- Users will provide valid and relevant data as per field requirements.
- The "Edit your response" feature is enabled, allowing users to revisit and modify their submitted responses.
- Edited responses will be processed and stored according to the default behavior of Google Forms (update or new entry).
- The system will be primarily accessed via desktop browsers, with basic support for mobile devices.

---

9. User Interface

The UI is clean and responsive, following Google's standard design - "Test A Form" e-form includes:

UI Element Type	**Description**
Title and Description	Displayed at the top of the form and in each section to guide the user.
Short Answer Field	Allows single-line text input.
Long Answer Field	Comment box for multi-line textual responses.
Dropdown Field	Enables selection from a predefined list of options.
Multiple-Choice Field	Presents single-option selection using radio buttons.
Checkbox Field	Allows multiple-option selection.
Date Field	Accepts input in date format via a date picker.
Time Field	Accepts time input via a time picker.
Numerical Validation	One or more fields require numeric input between two specified values.
Image Element	An embedded image for reference or visual context.
Section Breaks	The form is divided into two sections, guiding the user step-by-step.
Next Button	Navigates to the next section of the form.
Submit Button	Submits the form when all required fields are completed.
Confirmation Message	Shown upon successful submission: *"Your response has been recorded."*
Edit Response Link	Shown in the successful submission page
Sign-in Link (optional)	If present, clicking it redirects to the Google Sign-In page in a new tab.

10. Data Requirements

This section outlines the required data inputs, their types, whether they are mandatory, and any validation rules applied.

Field	**Type**	**Mandatory**	**Validation Rules**
Name	Short answer	Yes	Not empty
Email	Short answer	Yes	Valid email format
Are you a Tester?	Multiple Choice	Yes	Must select one option
Rate on scale of 1–10	Short answer	Yes	Must be a number between 1 and 10
Comments	Paragraph	No	Free text
Today's Date	Date	Yes	Valid date format (MM/DD/YYYY)
Check the boxes	Checkbox	No	At least one or more options can be checked
Time Check	Time	No	Valid time format (HH:MM)
Drop that list	Dropdown	No	Select one from list
Random image check	Image (Display)	N/A	Static image, no input required

---

11. System Behaviour

- Displays validation errors for incorrect inputs
- Prevents submission until required fields are valid
- Navigates between sections using "Next"
- Displays confirmation message after submission
- Stores responses in Google Sheets
- Provides "Edit your response" option post submission
- Redirects user to pre-filled form on edit
- Updates response upon re-submission

---

12. Dependencies

- Google Forms – Primary platform used for designing, hosting, and managing the e-form interface and validation logic.
- Google Sheets – Used for capturing, storing, and reviewing all submitted form responses.
- Internet Connectivity – Required for accessing the form, submitting responses, editing responses, and syncing data with Google Sheets.
- Web Browser – A compatible web browser (e.g., Chrome) is required to access and interact with the form.

13. Future Enhancements:

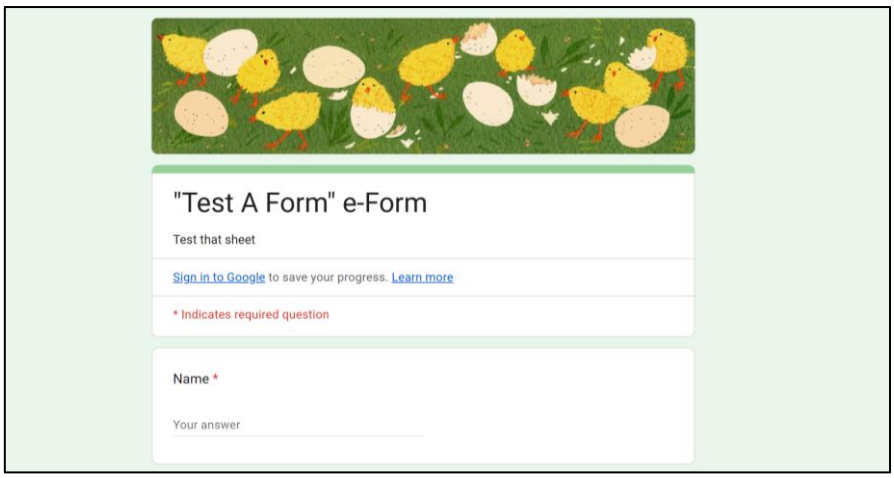
- API integration for validation
- Authentication-based submissions
- Reporting dashboards
- Data analytics integration
- CI/CD automation pipeline

---

14. URL and Version Updates

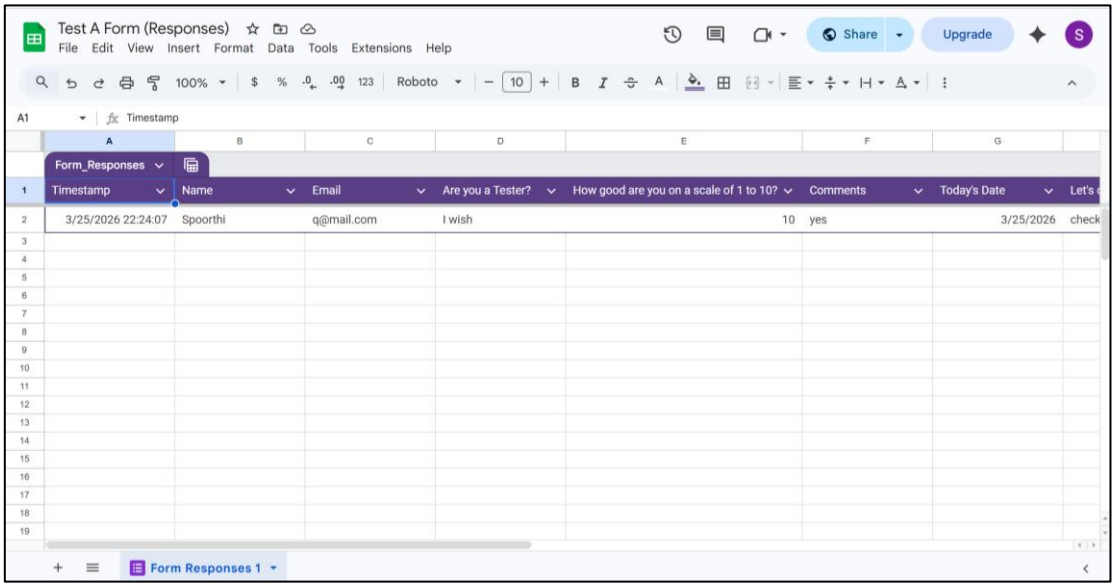
A. Page URL: Test A Form e-Form

- Landing Page:



B. Link to Response Google Sheet: Test A Form (Response)

- Sample response collection:



Version	Date	Author	Description of Changes
V-1.0	05-Aug-2025	Spoorthi	Initial draft created with project details
V-2.0	25-Mar-2026	Spoorthi	Enhanced FDD with refined scope, additional functional requirements, user stories alignment, and edit response functionality